

RVS UNIVERSITY

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

PROJECT REPORT

RESTAURANT BILLING SYSTEM

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in Electronics and Communication Engineering

Guided by

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CERTIFICATE

This is to certify that the project report entitled "Restaurant Billing System" has been successfully carried out by the below-mentioned students of IV B.Tech, Department of Electronics and Communication Engineering, RVS University, under the guidance of Naveen Sir. The work presented in this report is submitted as part of the academic requirements of the Bachelor of Technology programme.

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DECLARATION

We hereby declare that the project report entitled “Restaurant Billing System” is our original academic work carried out as a group project under the guidance of Naveen Sir, Department of Electronics and Communication Engineering, RVS University. The information presented in this report is prepared for academic purposes and has not been submitted elsewhere for the award of any degree or diploma.

Project Team

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ABSTRACT

The Restaurant Billing System is a console-based application developed using the C programming language to assist small restaurants in managing customer orders and generating bills. In a traditional restaurant setup, orders are often taken manually and bills are calculated by hand, which can lead to calculation errors, inconsistent pricing and delays during busy hours. The proposed system allows the restaurant staff to display the available menu, record the food items and quantities selected by a customer, calculate the cost of each item and generate the total bill automatically. The project demonstrates the practical application of core C programming concepts such as variables, data types, arrays, functions, loops, conditional statements, switch-case constructs and structures. The report presents the objectives, existing system, proposed system, functional modules, methodology, advantages, limitations and future scope of the Restaurant Billing System.

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1. INTRODUCTION

Restaurants handle a large number of daily activities such as displaying the menu to customers, taking orders, calculating the cost of items ordered and preparing the final bill. When these tasks are carried out manually, they can become time-consuming and prone to human error, particularly when a restaurant is handling multiple customers at the same time. A Restaurant Billing System provides a simple computerized environment in which the menu can be displayed, customer orders can be recorded, and the bill can be calculated automatically.

This project has been developed using the C programming language and is intended to demonstrate how fundamental programming concepts such as variables, arrays, functions, loops, conditional statements and structures can be applied to solve a real, practical problem. The system is designed as a console-based application suitable for a basic academic project in a B.Tech curriculum.

2. PROBLEM STATEMENT

In a traditional restaurant environment, orders are frequently written down on paper and bills are calculated manually using a calculator or by hand. This approach creates several difficulties, including:

- Manual recording of customer orders, which can be slow during busy periods.
- Calculation mistakes while adding item prices, especially for orders with several items.
- Repeated manual work for every new order, even when items and prices remain the same.
- Difficulty in maintaining consistent pricing across different orders and staff members.
- Time consumed in preparing a bill, leading to delays for customers.

These issues highlight the need for a simple, computerized billing solution that can reduce manual effort and calculation errors while remaining easy to use.

3. OBJECTIVES

- To display the restaurant menu along with the price of each food item.
- To allow the user to select food items from the menu.
- To accept the quantity required for each selected item.
- To calculate the item-wise cost based on quantity and price.
- To calculate the total bill amount for the complete order.
- To generate and display the final bill to the customer.
- To reduce manual calculation errors during the billing process.
- To demonstrate the practical use of core C programming concepts.
- To provide a simple and user-friendly console-based system.

4. EXISTING SYSTEM

In the existing traditional approach, restaurant staff record customer orders manually, usually on paper. The price of each item is looked up separately, and the total bill is calculated by manually adding the cost of every item ordered. This process is dependent on the accuracy and speed of the

staff member preparing the bill. Common drawbacks of this existing approach include manual order recording, manual price calculations, paper-based bills, human calculation errors, time consumption during peak hours and difficulty in managing orders that contain a large number of items.

5. PROPOSED SYSTEM

The proposed Restaurant Billing System improves the basic billing workflow by automating the calculation process using a C program. The general process followed by the system is as follows:

- The system displays the available menu along with item prices.
- The user selects a food item from the displayed menu.
- The user enters the quantity required for the selected item.
- The system calculates the cost of the selected item based on quantity and price.
- The selected item is added to the current order.
- The user can continue selecting additional items if required.
- Once ordering is complete, the system calculates the final total amount.
- The system displays or generates the final bill for the customer.

This workflow is implemented as a console-based C application and does not depend on any external database or network service.

6. SYSTEM MODULES

The system can be understood in terms of the following logical modules, each of which represents a distinct part of the program's functionality rather than a separate software component.

Menu Display Module

Displays the list of available food items along with their prices to the customer or staff member.

Order Processing Module

Accepts the customer's food item selections and the quantity required for each item.

Bill Calculation Module

Calculates the cost of each selected item based on quantity and price, and computes the subtotal for the order.

Bill Generation Module

Displays the final order summary, including individual item costs and the total bill amount.

Input Handling Module

Manages basic user input, menu selection and validation of entered values.

7. FUNCTIONAL REQUIREMENTS

- The system shall display the restaurant menu with item names and prices.
- The system shall accept the customer's food item selection.
- The system shall accept the quantity required for each selected item.

- The system shall calculate the price of each selected item.
- The system shall calculate the total bill amount for the order.
- The system shall display the complete order details before generating the bill.
- The system shall generate and display the final bill.
- The system shall allow multiple items to be added to a single order.

8. NON-FUNCTIONAL REQUIREMENTS

- Usability: The system should be simple and easy for restaurant staff to operate.
- Accuracy: Bill calculations should be correct and consistent.
- Reliability: The system should function correctly for repeated orders during a session.
- Simple Interface: The console interface should be clear and easy to follow.
- Fast Calculation: The billing calculation should be performed quickly.
- Maintainability: The C source code should be organized so that it can be understood and modified.
- Portability: The program should be portable across systems that support standard C compilation.

9. SYSTEM DESIGN AND WORKFLOW

The overall workflow of the Restaurant Billing System can be summarized as follows:

1	START
2	Display Restaurant Menu
3	Select Food Item
4	Enter Quantity
5	Calculate Item Cost
6	Add Item to Order
7	Check if More Items are Required
8	If YES → Select Another Item
9	If NO → Calculate Final Bill
10	Display Final Bill
11	END

This workflow reflects a straightforward, sequential program flow that is consistent with a console-based C application built using loops and conditional statements.

10. METHODOLOGY

The project was developed using a simple, structured programming approach appropriate for a C-based academic project. The main stages followed during development were:

- Requirement identification: understanding the basic billing needs of a small restaurant.

- Menu and order design: deciding the food items, prices and structure of an order.
- C program implementation: writing functions for menu display, order entry and bill calculation.
- Input processing: handling user input for item selection and quantity.
- Bill calculation: implementing logic to compute item-wise and total costs.
- Testing: verifying the program with different orders to check correctness of calculations.
- Final output generation: displaying the completed bill to the user.

11. ADVANTAGES

- Faster billing compared to manual calculation.
- Reduced calculation errors in the final bill.
- Easy handling of the menu and item prices.
- Simple operation suitable for everyday use.
- Accurate calculation of the total order amount.
- Reduced manual effort for restaurant staff.
- Easy to understand and operate, even for a new user.
- Useful for demonstrating practical application of core C programming concepts.

12. LIMITATIONS

- The system uses a console-based interface without a graphical user interface.
- There is no database; order and menu data are handled within the running program.
- The system does not include online payment integration.
- Data is not stored permanently once the program session ends.
- The system is suitable mainly for basic restaurant billing needs.
- The C program needs to be compiled and executed locally on a computer.

13. FUTURE SCOPE

The current project can be extended in the future with additional features, including:

- A graphical user interface for easier interaction.
- Database integration for permanent storage of orders and menu items.
- Maintenance of customer records for repeat visits.
- Basic inventory management for tracking available stock.
- Support for digital payments.
- Automatic receipt printing.
- User authentication for staff login.
- A web or mobile version of the application.
- Generation of sales reports over a period of time.
- An administrative dashboard for restaurant owners.

These enhancements are proposed as possible future improvements and are not part of the current implementation.

14. CONCLUSION

The Restaurant Billing System demonstrates how the C programming language can be applied to solve a practical, real-world problem faced by small restaurants. Through this project, concepts such as menu management, order processing, bill calculation and basic data handling have been implemented using variables, arrays, loops, conditional logic, functions and structures. The project shows that even a relatively simple console-based application can meaningfully reduce manual effort and calculation errors in restaurant billing, while also serving as a practical demonstration of core programming concepts learned during the course.

15. REFERENCES

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